

Trajectory-Space Omega: A Research Note and Empirical Roadmap

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Abstract

This note proposes a trajectory-space branch of the Omega research program. The current Omega artifact records a quotient-relative COM fiber-transport witness in a toy multifield substrate. The trajectory-space branch asks whether viable propagation can be detected more directly from the measure and temporal structure of viable trajectories, before choosing a macro-fiber quotient.

The proposed profile tracks viable trajectory mass, hazard structure, conditional temporal dependence, recoverability, trajectory concentration, and multifield component dependence. The purpose is not to replace the COM witness immediately, but to test whether the COM construction is an operational discretization of a deeper trajectory-space signature. The empirical roadmap begins with a small smoke test of viability mass and conditional temporal dependence, then proceeds to component preservation without fibers and direct comparison against the COM witness.

Status and Scope

This document is a planning note for a parallel research branch. It does not revise the current COM fiber-transport artifact. The COM witness remains the best-audited quotient-relative toy signature. The trajectory-space branch is intended to test whether that witness is part of a deeper quotient-light formulation based on viable trajectory measures and temporal dependence.

Introduction

Purpose

The current Omega artifact records a quotient-relative COM fiber-transport witness in a toy multifield substrate. This note proposes a parallel branch: a trajectory-space formulation of Omega that attempts to reduce dependence on hand-chosen macro-quotients.

The guiding question is:

Can viable propagation be detected directly from the measure and temporal structure of viable trajectories, before choosing a macro-fiber quotient?

This branch does not replace the COM witness. It tests whether the witness is an operational discretization of a deeper trajectory-space signature.

Core Hypothesis

Let μ_s be the trajectory measure induced by initial state s and dynamics P , and let:

$$\mathcal{F}_T(s) = \{\tau_{0:T} : \tau_0 = s, V(\tau_{0:T}) = 1\} \quad (1)$$

be the set of viable trajectories over horizon T .

A trajectory-space Omega profile should track not only viable mass,

$$p_T(s) = \mu_s(\mathcal{F}_T(s)), \quad (2)$$

but also whether viable trajectory structure propagates through time:

$$I_t^V = I(\tau_{0:t}; \tau_{t:T} \mid V(\tau_{0:T}) = 1). \quad (3)$$

The working hypothesis is:

Omega-like systems preserve viable trajectory mass and temporal dependence without collapsing into brittle, component-erasing, or single-path regimes.

Candidate Trajectory-Space Profile

The proposed trajectory-space profile is:

$$\Omega_{\text{traj}}(s, T) = (p_T, h_T, I_t^V, R_T, Q_T, C_{AB}, K_T), \quad (4)$$

where:

- p_T is viable trajectory mass;
- h_T is the hazard or first-failure profile;
- I_t^V is conditional temporal dependence under viability;
- R_T is recoverability under perturbation;
- Q_T is trajectory concentration or collapse;
- C_{AB} is component-preserving conditional dependence in multifield systems;
- K_T is comparison to product, shuffled, or time-shuffled null baselines.

This profile is intentionally not a single scalar. Different failure modes appear in different coordinates.

Relation to the COM Witness

The current COM witness is:

$$\Omega_{\text{witness}} = \mathcal{P}_T^{\kappa_{\text{COM}}}(S_{\text{toy}}). \quad (5)$$

The trajectory-space branch asks whether this witness corresponds to a quotient-light or quotient-free trajectory profile:

$$\mathcal{P}_T^{\kappa_{\text{COM}}} \quad \text{as an operational discretization of} \quad \Omega_{\text{traj}}. \quad (6)$$

Possible outcomes:

1. COM and trajectory-space profiles agree.
2. Trajectory-space detects propagation missed by COM.
3. COM detects quotient-specific structure not visible in trajectory space.
4. Both are weak outside the current substrate.

Each outcome is informative.

Milestone T1: Viable Trajectory Measure

Question. Does viable trajectory mass distinguish open, trapped, noisy, rigid, and COM-positive regimes?

Observable.

$$p_T(s) = \mu_s(\mathcal{F}_T(s)). \quad (7)$$

Metrics.

- viable mass p_T ;
- survival curve over time;
- first-failure hazard profile;
- endpoint concentration;
- trajectory concentration.

Initial substrates.

- single-field open world;
- sink/trap world;
- rigid attractor;
- noise swamp;
- multifield F, T attractive corridor.

Success criterion. Trajectory-space viability should separate traps and rigid collapse from robust viable regimes without relying on a quotient.

Milestone T2: Conditional Temporal Dependence

Question. Given viability, does early trajectory structure remain informative about later trajectory structure?

Observable.

$$I_t^V = I(\tau_{0:t}; \tau_{t:T} \mid V(\tau_{0:T}) = 1). \quad (8)$$

Practical estimators.

- Gaussian mutual information approximation;
- predictive regression score;
- discretized coordinate-bin mutual information;
- time-shuffled null comparison.

Metrics.

- conditional temporal MI by segment split;
- MI degradation slope over horizon;
- predictive future score;
- time-shuffled MI delta;
- concentration-adjusted MI.

Success criterion. Omega-like regimes should preserve conditional temporal dependence above time-shuffled and product baselines, without collapsing into a single deterministic path.

Milestone T3: Component Preservation Without Fibers

Question. Can multifield component preservation be detected without macro-fibers?

For fields A and B , test:

$$I(\tau_{0:t}^A; \tau_{t:T}^A \mid \tau_{0:T}^B, V(\tau_{0:T}^{AB}) = 1), \quad (9)$$

and symmetrically for B .

Metrics.

- A self-future information conditioned on B and viability;
- B self-future information conditioned on A and viability;
- component information balance;
- component erasure proxy;
- product and shuffled baselines.

Success criterion. A multifield Omega-like regime should preserve joint viable propagation without erasing either component's conditional future information.

Milestone T4: COM-vs-Trajectory Agreement

Question. Does the known COM witness correspond to a trajectory-space propagation signature?
Compare.

- COM viable propagation index;
- viable trajectory mass;
- conditional temporal MI;
- component conditional MI;
- trajectory concentration;
- recoverability under perturbation.

Conditions.

- F, T attractive corridor;
- $\alpha \in \{0.45, 0.50, 0.525\}$;
- $T \in \{900, 1500, 2400\}$;
- product, shuffled, and time-shuffled baselines.

Interpretation.

If COM and trajectory-space signatures agree, COM may be a finite-state operational witness of a deeper trajectory-space object. If they diverge, the difference clarifies whether the COM result is quotient-specific or whether trajectory-space reveals additional structure.

Milestone T5: Perturbation and Horizon Robustness

Question. Does the trajectory-space signature survive the same perturbations that COM survived in Probe 10?

Perturbations.

- noise level;
- potential shape;
- time discretization;
- sink threshold;
- initial condition shifts.

Success criterion. The trajectory-space signature should degrade gracefully under perturbation and show interpretable failure modes. It does not need to survive all perturbations.

First Empirical Probe

The first empirical step should combine T1 and T2 as a smoke test.

Probe name.

Probe T1/T2: Trajectory-Space Viability and Temporal Dependence Smoke Test

Budget.

- soft limit: 90 minutes;
- hard limit: 2 hours;
- workers: 18;
- trajectories: 5000;
- seeds: 100;
- bootstrap repeats: 300.

Conditions.

- single-field open, trap, rigid, noise controls;
- multifield F, T attractive corridor;
- $\alpha \in \{0.50, 0.525\}$;
- $T \in \{900, 1500\}$.

Outputs.

- `summary.json`
- `viability_profile.csv`
- `hazard_profile.csv`
- `temporal_dependence.csv`
- `trajectory_concentration.csv`
- `baseline_comparisons.csv`
- `estimator_report.csv`

Primary readout.

Does trajectory-space temporal dependence distinguish viable propagation from mere survival, traps, noise, and COM-positive multifield regimes?

Interpretive Guardrails

The trajectory-space branch should preserve the same three principles:

- **Minimal:** begin with viability mass and conditional temporal dependence before adding complex estimators.
- **Principled:** use null baselines, perturbations, and collapse diagnostics from the start.
- **Revelatory:** allow the trajectory-space profile to agree with, refine, or contradict the COM witness.

The branch should not be treated as automatically superior to the quotient route. Continuous trajectory-space methods still require choices of metric, density estimator, discretization, and representation. The goal is to determine whether those choices are less load-bearing than the explicit macro-quotient.

Strategic Role

This branch gives the project a way to test whether Omega is better grounded as:

1. a quotient/fiber transport object;
2. a trajectory-measure viability object;
3. an information-theoretic temporal propagation object;
4. or a structured profile combining all three.

The current expectation is that the mature theory may use trajectory-space viability and information flow as a backbone, with quotient/fiber witnesses as finite-state operational projections.

Near-Term Decision Point

After Probe T1/T2, decide:

- If trajectory-space metrics separate the known regimes cleanly, proceed to T3 component preservation without fibers.
- If trajectory-space metrics reduce to survival only, strengthen the temporal-dependence and recoverability measures.
- If trajectory-space metrics contradict the COM witness, run T4 to map the disagreement.
- If estimators are unstable, simplify to predictive surrogates before attempting full mutual information.

Project Artifacts

This note is based on the current Omega project state after Probes 09–12 and the COM fiber-transport witness artifact. Formal external references to viability theory, information theory, active inference, and option-value theory should be added in later drafts if this branch develops into a full paper.

- docs/PUBLIC_RESULTS_INDEX.md
- probe_09_robust_fiber_reachability_results/
- probe_10_com_viable_propagation_robustness_extended_results/
- probe_10_com_targeted_fragility_refinement_results/
- probe_11_learned_predictive_kappa_revised_results/
- probe_12_batch_results/